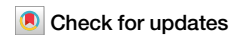


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In situ observations reveal continental-scale warming, oxygen decline, and eutrophication in U.S. estuaries



Kaitlin L. Reinl¹ ✉, Robert P. Dunn^{2,3}, Kimberly A. Cressman⁴, Theophilos Collins⁵, Jennifer L. DeBose^{6,7}, Carl T. Friedrichs⁸, Jason S. Goldstein⁹, Alicia R. Helms¹⁰, Christopher Kinkade¹¹, Julie L. Krask², David B. Parrish⁸, Hannah N. Nicklay¹², Justin T. Ridge¹³, Denise M. Sanger¹⁴, Jacob A. Cianci-Gaskill¹⁵, Nicole G. Dix^{16,17}, Thomas M. Grothues¹⁸, Steven E. McMurray¹⁵ & Christopher Peter¹⁹

Estuaries are sentinel ecosystems for understanding the impacts of climate and land-use change, but availability of in situ data for continental-scale comparative studies is rare. Here, we document the status of water quality in 29 U.S. estuaries across multiple climate zones using in situ data to detail decadal-scale trends in key environmental parameters and their drivers. Our analysis assembles more than 250 million data points from over the past two decades (2002–2022), providing direct evidence of widespread eutrophication and warming in estuaries, with significant increases in chlorophyll-*a* and water temperature at approximately two-thirds of sites, and oxygen declines at half of sites. Changing nutrient concentrations, not temperature, appear to be the primary driver of increases in chlorophyll-*a* at the continental scale, while increasing temperature and chlorophyll-*a* are the best predictors of declining oxygen concentrations. These analyses reinforce that changes in coastal ecosystem dynamics are widespread and complex, driven by multiple interacting factors that require large-scale, standardized data collection to disentangle.

Estuaries, situated at the critical interface between terrestrial and aquatic ecosystems, are ecologically and economically valuable environments that are increasingly threatened by anthropogenic and climate-driven stressors¹. While estuary characteristics vary considerably across geographic settings, they are consistent in their ecological role of serving as critical nursery habitats, fostering high levels of productivity, and facilitating active biogeochemical cycling largely driven by mixing of chemically-distinct water masses as well as nutrient inputs from surrounding watersheds^{2–4}. This productivity and habitat provision equates to billions of dollars in ecosystem services to coastal economies, particularly for fisheries, recreation, and tourism industries⁵. However, the location of estuaries at the land-water interface exposes them to both terrestrial (e.g., land use change, runoff) and aquatic (e.g., sea level rise, ocean acidification, harmful algal blooms) stressors^{6–9}.

Eutrophication is a common threat to the water quality of coastal ecosystems^{10,11}. During eutrophication, increased nutrient inputs trigger overgrowth of phytoplankton, in some cases resulting in harmful algal blooms (HABs), which some studies show have been increasing in extent and frequency over previous decades¹². Decomposition of phytoplankton

aggregations following these blooms can significantly lower oxygen levels through microbial uptake during decomposition, causing hypoxia (defined herein as dissolved oxygen [DO] <2 mg/L) and anoxia (complete oxygen depletion). These low-oxygen events reduce estuarine nursery function and negatively impact fish populations¹³ and benthic macroinvertebrate communities¹⁴. Further, hypoxia can alter sediment redox conditions, leading to internal nutrient loading of dissolved nutrients back into the water column, creating a self-sustaining mechanism for subsequent phytoplankton blooms^{7,12}. Increased phytoplankton biomass also reduces light penetration in the water column and to the benthos, triggering negative ecosystem-level impacts^{15,16}. Coastal eutrophication has previously been estimated to cost \$100 million annually for the United States, and \$1 billion for European Union nations, due to public health concerns, lost recreational opportunities, and degraded commercial fisheries¹⁷. Coastal eutrophication and hypoxia are longstanding problems in the U.S. and despite regulatory efforts dedicated to combating these issues, they continue to pose significant ecological, economic, and public health risks.

Estuaries are considered sentinel ecosystems for understanding the impacts of climate change^{18,19}, storm events^{20,21}, and land-use change^{22,23}, but

A full list of affiliations appears at the end of the paper. ✉e-mail: reinlk@rpi.edu

in situ, continental-scale comparative studies are relatively rare. Observations of environmental conditions and associated ecological processes from individual estuaries^{24–27} or regional assessments^{19,28}, while highly informative, have limited broad-scale application due to the diversity in estuarine attributes across continental scales. Large-scale assessments (e.g., national, continental, and global) of estuarine eutrophication have focused on qualitative syntheses or remotely sensed data^{12,29,30}. While these data types are critical to estimate spatial patterns at broad scales, they can suffer from observation bias and misinterpretation, coarse or limited resolution and quality (e.g., gaps in time-series coverage), mismatch in time and space leading to missed events, pixel bleed and bottom reflectance in small or shallow estuaries³¹, and often lack of finer-scale ground-truthing. Alternatively, in situ data collected at high spatial and temporal resolutions provides objective, replicable, and high-integrity measurements that enable the characterization of disruptive events and sustained trends in these systems.

The U.S. National Estuarine Research Reserve System (NERRS) System-Wide Monitoring Program (SWMP) was designed to quantify short-term variability and long-term ecosystem change within a network of 30 coastal locations (i.e., Reserves) around the United States, including Hawaii, Alaska, and Puerto Rico^{6,32–34}. SWMP data have previously been utilized to characterize spatial and temporal patterns in water quality^{35–37}, investigate the importance of autotrophic and heterotrophic processes^{38,39}, understand mechanisms regulating phytoplankton biomass⁴⁰, and quantify the effects of eutrophication¹³ and ocean acidification⁴¹ on water quality. Given the range of geomorphologies and climatic conditions of estuaries within the NERRS, application of standardized methods across all monitored sites, and time span of sample collection (i.e., ~20 years included in this analysis), the SWMP is uniquely positioned to address large-scale spatial and temporal questions regarding the status and recent trajectory of estuarine conditions in the United States^{33,42}. Here, we document the status of estuarine ecosystems across the NERRS using standardized in situ monitoring data, detail decadal-scale trends in key environmental parameters, and investigate factors contributing to trends in eutrophication and oxygen declines. Our analysis brings together >250 million data points collected from 129 stations from 2002 to 2022, providing a unique investigation of in situ estuarine conditions at a continental scale.

Results and discussion

Status of water quality in U.S. estuaries

Across 129 monitoring sites located in 29 of 30 National Estuarine Research Reserves (Supplementary Data 1), cluster analysis revealed four types of estuarine sites exhibiting shared biochemical characteristics (Figs. 1 and 2, Supplementary Data 2, 3). Median water temperature ranged from 6.2 to 29.3 °C across all clusters, and specific conductivity (SpCond) ranged from 0.2 to 69.9 mS/cm, including predominant fresh and saltwater influences, with median turbidity ranging from 1.0 to 55 NTU. Median DO fell between 2.4 and 10.8 mg/L with an average median value by cluster of 7.4 mg/L. The pH among all 129 sites was somewhat variable, with a mode of 6.3 to 8.3. Chlorophyll-*a* medians by cluster ranged from 0.2 to 30.5 µg/L, and nitrite + nitrate (NO_x), ammonium (NH₄⁺), and orthophosphate (PO₄³⁻) ranged between 0.0013–3.1, 0.002–0.44, and 0.0008–0.23 mg/L, respectively. Sites in Cluster A exhibited the highest average median dissolved inorganic nitrogen (DIN; NO_x = 0.30 mg/L, NH₄⁺ = 0.051 mg/L), lowest pH (7.34), and lowest specific conductivity (7.9 mS/cm; SpCond is highly correlated with salinity in this dataset, Supplementary Fig. 1), often associated with the most freshwater influenced sites. Cluster A sites also had the highest mean turbidity (18 NTU) and chlorophyll-*a* (chl-*a*; 6.5 µg/L), suggesting higher suspended particle concentrations (including organic detritus). Cluster B sites experienced higher SpCond (39.8 mS/cm), pH (7.9), and temperature (24.9 °C), and lower DIN than the other clusters (NO_x = 0.0076 mg/L, NH₄⁺ = 0.015 mg/L), suggesting more influence from warm, marine waters compared to other sites. Cluster C sites had the lowest average water temperatures (12.5 °C) accompanied by the highest DO (8.9 mg/L), and a large

range of SpCond (0.26–69.9 mS/cm). Importantly, Cluster C includes many sites where warming trends were strongest, including in the northeast U.S. (Fig. 3). These sites also tended to have the lowest mean chl-*a* concentrations (1.69 µg/L) as well as low turbidity (2.0 NTU), suggesting relatively low organic and inorganic particle concentrations. Cluster D somewhat mirrored Cluster B with relatively higher water temperatures (20.7 °C) and low mean DO concentration (6.2 mg/L), but these sites are distinguished by their high mean ammonium (NH₄⁺; 0.049 mg/L) and orthophosphate (PO₄³⁻; 0.038 mg/L) concentrations. We also evaluated any effects of sensor depth on Cluster distribution and only observed a weak relationship with Cluster C, attributed to three deeper sites greater than 5 meters which were not included in the long-term trend analysis (Supplementary Figs. 2 and 3). Though the four identified clusters are distinct at their centroids, sites near the edges may exhibit characteristics of other clusters, and in reality, fall somewhere on a continuum of characteristics associated with a given cluster⁴³ (Fig. 2 and Supplementary Fig. 4).

In addition to being biochemically distinct, the clusters exhibit geographic patterns (Fig. 2). Clusters A and C generally appear north of the 37 °N latitude position, and Clusters B and D are more often found south of this latitude position. Fifteen estuaries had sites that fell within only one cluster, while eleven estuaries had two of the four clusters represented. Finally, three estuaries (Great Bay, NH, GRB; Jacques Cousteau, NJ, JAC; and Chesapeake Bay, VA, CBV) have sites represented in Clusters A, C, and D. These patterns seem to be associated with latitude position, which would influence temperatures and subsequently, DO concentrations. Other factors that likely contribute to patterns in cluster grouping are phytoplankton biomass, indicated by chl-*a* concentrations, and turbidity, which is somewhat linked to chl-*a* concentrations ($R^2 = 0.24$ and $p < 0.05$; Supplementary Fig. 5). These four clusters represent distinct water quality regimes that reflect the water quality status of these 129 sites, offering a useful framework for interpreting spatial patterns in estuarine nutrient dynamics, salinity gradients, and trophic state across the national network.

Trends in estuarine conditions

We analyzed trends for 99 study sites and found that the majority of sites across 27 U.S. estuaries are experiencing significant, increasing trends in chl-*a* (70%) and water temperature (66%), whereas half of sites (49%) exhibited significant reductions in DO concentrations (Fig. 3 and Supplementary Data 4). Our findings are consistent with evidence of widespread warming, eutrophication, and hypoxia from other recent studies of coastal environments^{11,44}. Further, the overall increase in temperature exhibited by most estuaries almost assuredly played a key role in reducing DO levels (as documented in the section below). Despite the proportion of sites exhibiting oxygen declines, only eight sites exhibited significant increasing trends in the proportion of time under hypoxic conditions (DO < 2 mg/L; APA-EB, SAP-DC, TJR-BR, TJR-OS, ELK-PW, HUD-TN, OWC-DR, and SFB-GC; Slopes are reported in Supplementary Data 4 and reserve/station codes are in Supplementary Data 1), suggesting that while trends in DO are generally negative, water quality conditions have not yet deteriorated to the degree to which significant negative ecological impacts related to low DO concentrations would be observed.

Nutrient trends (NO_x, NH₄⁺, and PO₄³⁻) showed variability in their directionality, both increasing and decreasing across U.S. estuaries, though the direction of PO₄³⁻ and NH₄⁺ trends were generally consistent in their direction across sites (i.e., positively related; Supplementary Fig. 6). However, the relationships between NO_x and PO₄³⁻, and NO_x and NH₄⁺, were more variable, showing no clear relationships (Supplementary Figs. 7 and 8), highlighting the complex interactions between these nutrient forms. These nutrients are highly sensitive to many unique biogeochemical changes at the local scale⁷, making establishing clear relationships between them at the continental scale exceptionally difficult.

The trends in temperature, chl-*a*, and DO concentrations were not unique to a particular cluster (Fig. 4), although there are some notable regional patterns (Fig. 3). Many of the highest rates of warming and steepest

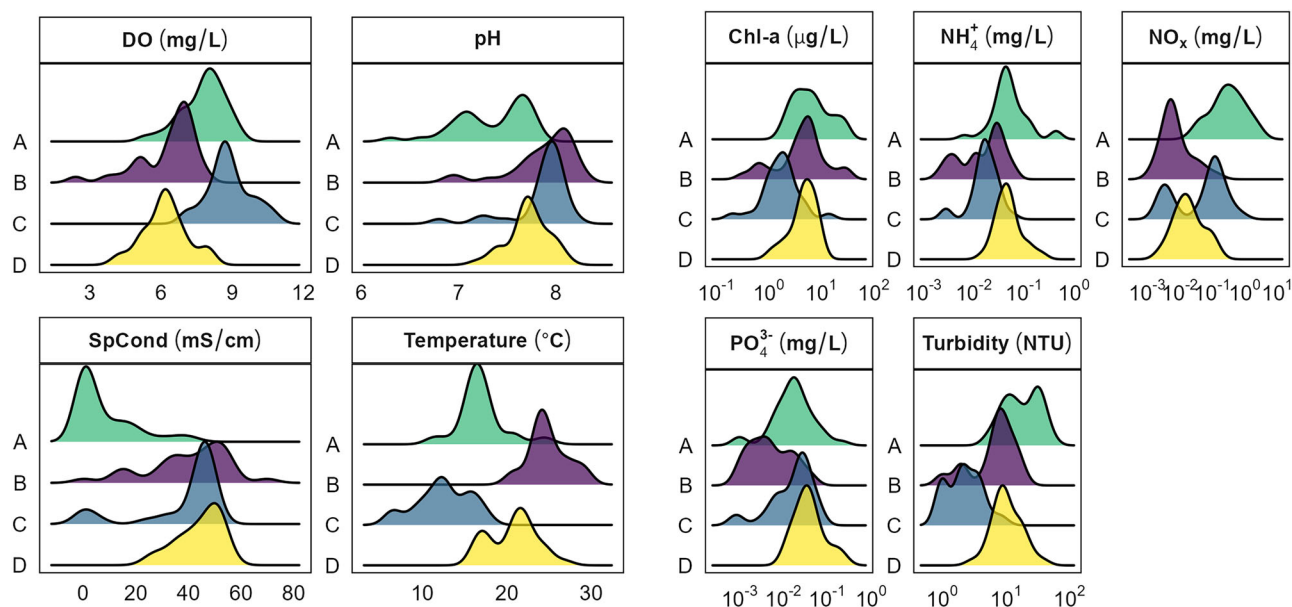


Fig. 1 | Distribution of median water quality and mean nutrient characteristics in each cluster.

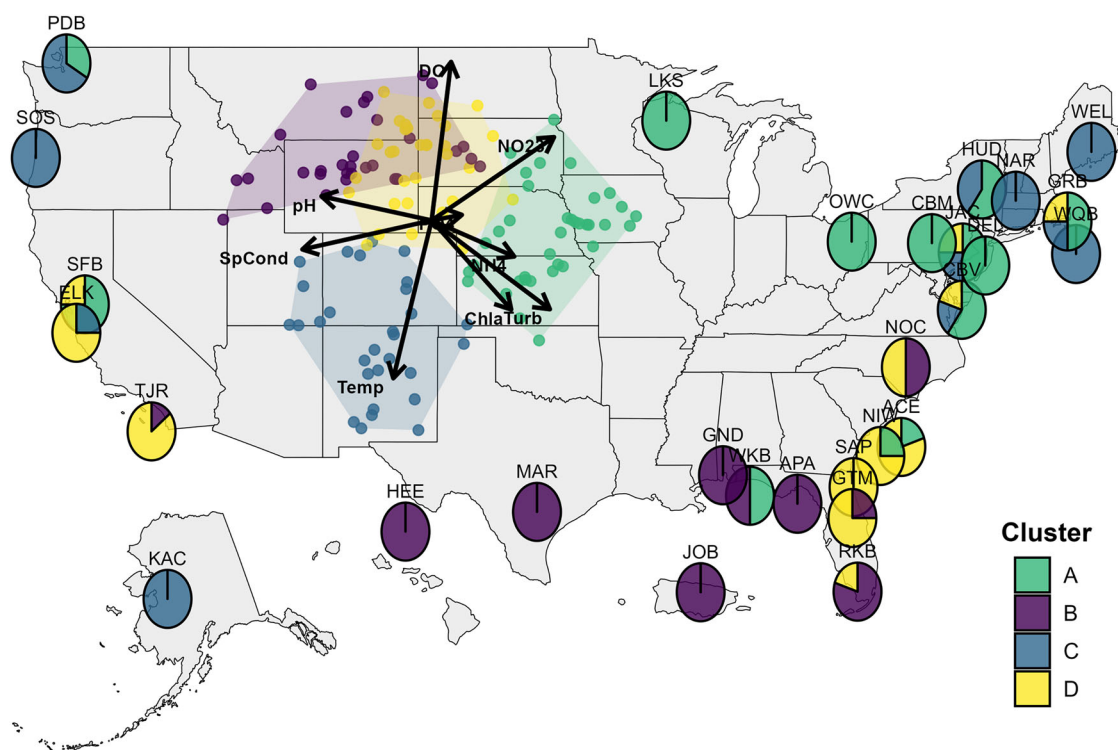


Fig. 2 | Geographic distribution of sites associated with each cluster. The PCA shows PC1 (32.4%) on the x-axis and PC2 (25.9%) on the y-axis with clusters. A more detailed version that includes PC 3 can be found in Supplementary Fig. 4.

Three letter codes represent the abbreviated names of the reserves (see Supplementary Data 1) and corresponding pie charts show how the different stations within each reserve fit into the four clusters.

declines in oxygen levels were observed at sites in the Northeast U.S. (WEL, GRB, WQB, and NAR; all reserve and station codes in Supplementary Data 1) with rates of temperature increase ranging from 0.06–0.08 °C/year, similar to other recent estimates of warming in this region⁴⁵. Other locations with similar rates of change in temperature include sites at the Tijuana (TJR) and Sapelo Island (SAP) NERR, both located at more southerly latitudes (~32 °N). Lower rates of temperature increase tended to be common further north on the U.S. Pacific coast, which may be due to the modulating effects of ocean upwelling, damping local atmospheric and estuarine warming⁴⁶.

Potential drivers of estuarine eutrophication and oxygen declines

Temporal changes in chl-a concentration across U.S. estuaries were best predicted by corresponding site-level trends in NH_4^+ , PO_4^{3-} , and turbidity, with a relative importance of 0.98, 0.95, and 0.90, respectively (Fig. 5 and Supplementary Data 5). It is important to note that because nearly all chl-a trends were positive, as the slope of NH_4^+ becomes more negative, the magnitude of the chl-a trend becomes even more positive (Fig. 6). Similarly, increases in chl-a occur when the trend in both turbidity and PO_4^{3-} are both positive and negative, but the highest rate of chl-a increases occur when

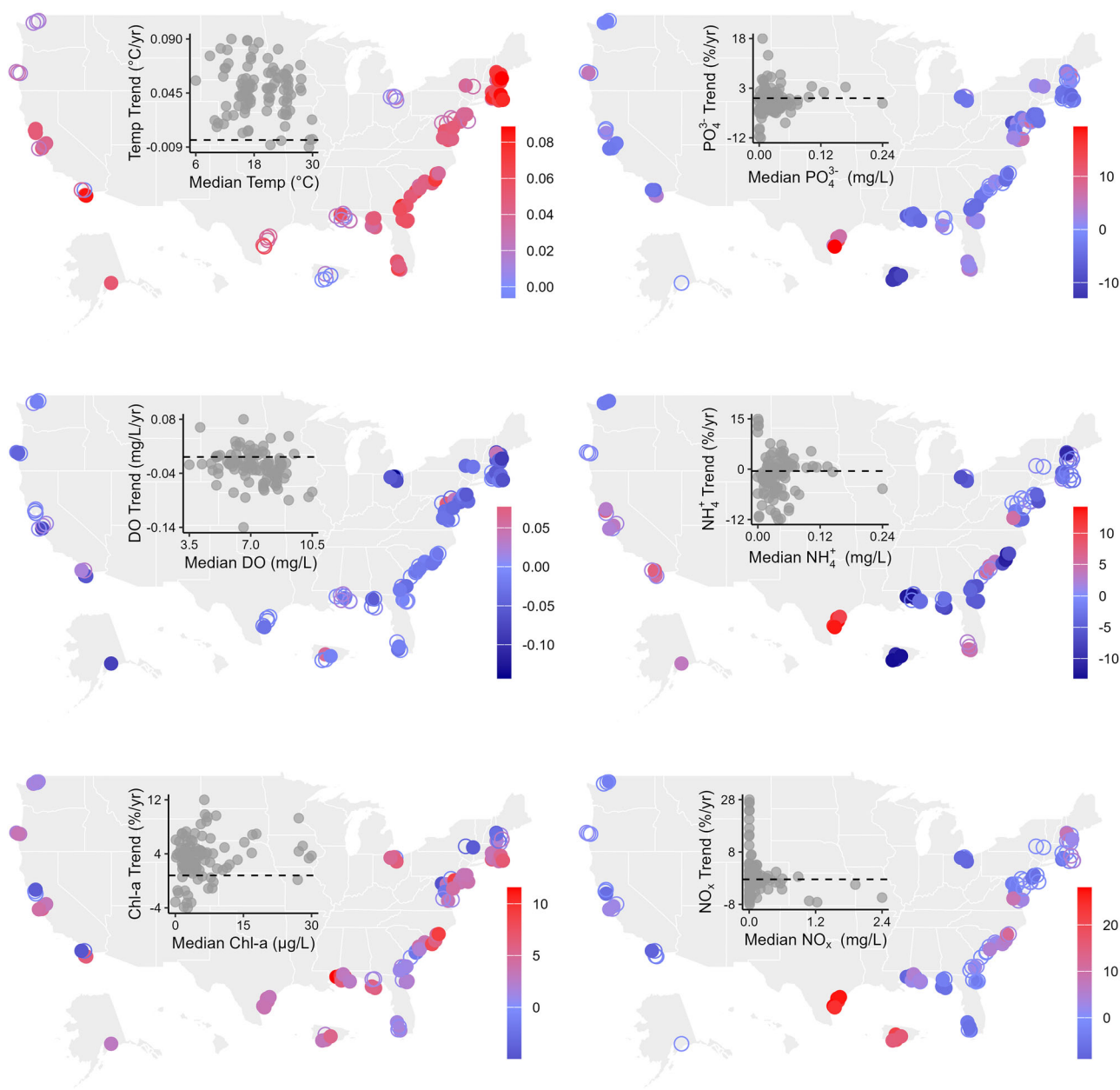


Fig. 3 | Geographic patterns of slopes for trends in six water quality parameters: orthophosphate (PO_4^{3-}), ammonium (NH_4^+), nitrite+nitrate (NO_x), dissolved oxygen (DO), temperature, and chlorophyll-a (chl-a) concentration. Filled circles on the map denote statistically significant slopes ($p < 0.05$). The color bar legend corresponds to slope values plotted on the map with units reported as % per year,

with the exception of DO (mg/L/yr) and Temp ($^{\circ}\text{C}/\text{yr}$), where blue colors indicate more negative slopes and red indicates more positive slopes. Within each panel, the scatterplot shows the relationship between median and trend slope values across all sites.

trends in turbidity and PO_4^{3-} are both positive (Fig. 6). Of the 165 chl-a models with $\Delta\text{AICc} < 4$ (AIC: Akaike Information Criterion, AICc: corrected AIC), these variables (in the order presented above) appeared in 163, 157, 145, and 109 of the 32,768 candidate models. The marginal R^2 for the global model (i.e., all fixed effect predictors⁴⁷) was 0.19, while the conditional R^2 (i.e., complete model with fixed and random effects) was 0.71, meaning that site specific factors for each of the 27 reserves (i.e., allowing intercept to vary for sites in each reserve to allow for baseline differences) explained an additional 50% of the variance in chl-a trends across the 99 long-term monitoring sites. When only including the top four predictors, the marginal R^2 was 0.10 and the conditional R^2 was 0.60. Importantly, although we assume our sites are well mixed, in situ water samples in the SWMP dataset are typically collected just off the benthos ($<0.5\text{ m}$); therefore, we are

conservatively estimating chl-a and the magnitude of the trends in chl-a could actually be larger.

Trends in DO concentration were best predicted by trends in temperature and chl-a, as well as median SpCond and chl-a. DO trends were inversely related to temperature and chl-a trends, and positively related to median SpCond and chl-a. More specifically, more positive trends in temperature and chl-a predicted larger decreases in DO as nearly all DO trends were negative (Fig. 6). The relative importance of all four of these predictors was 1.00 (Fig. 5 and Supplementary Data 6). Temperature trends, median SpCond, and chl-a trends appeared in all 139 models with $\Delta\text{AICc} < 4$, while the log of median chl-a (trophic status) appeared in 137 out of 65,536 candidate models. The global model (including all predictors) yielded an adjusted R^2 of 0.12, while the top four predictors yielded an adjusted R^2 of

Fig. 4 | Trend slopes for chl-a, dissolved oxygen, and water temperature by cluster.

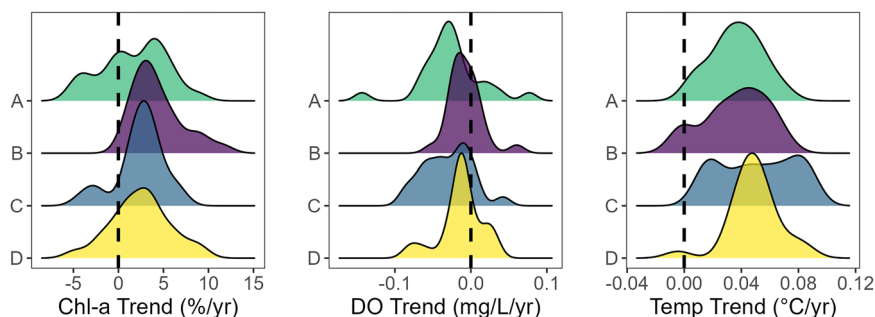
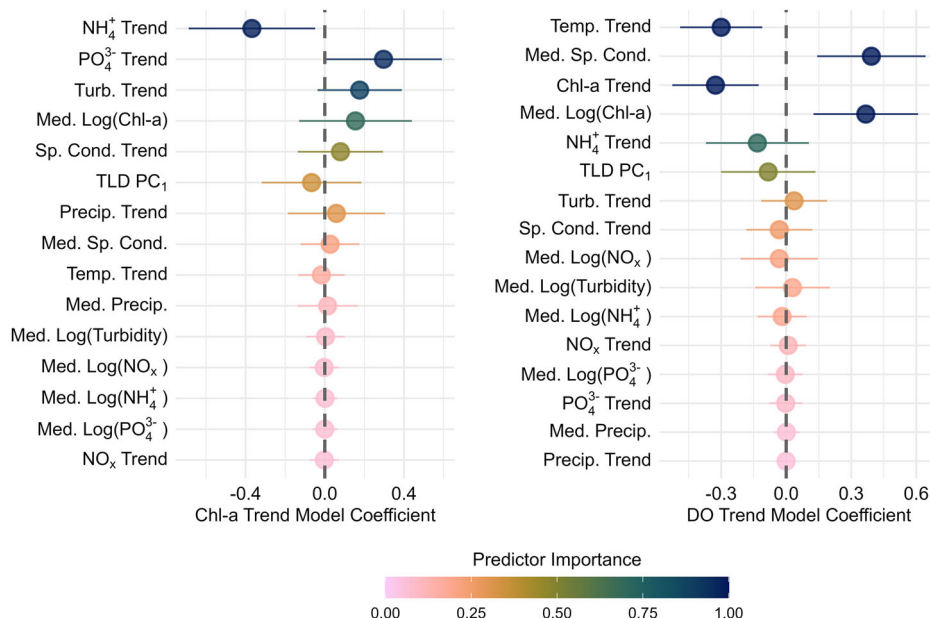


Fig. 5 | Standardized coefficients based on model averaging. Colors represent relative importance of predictor variables; note that predictors are not presented in the same order between left and right panels. For chl-a, 165 of the original 32,768 potential model subsets were retained based on our delta AICc threshold of 4. For DO, 139 of the original 65,536 subsets were retained and used for coefficient estimation via model averaging. Temperature/Latitude/Depth (TLD) PC1 indicates the first principal component of the PCA analysis for median temperature, DO, and latitude as model predictors (see methods).



0.14. No marginal R^2 is reported for this model as no random effect was present.

Our results suggest that nutrients, specifically NH₄⁺ and PO₄³⁻, are more strongly associated with trends in phytoplankton biomass than water temperature, while NO_x trend was at the bottom of the predictor pool. Typically, nitrogen is identified as the most common limiting nutrient in coastal systems⁷, but this is not a spatiotemporal absolute. Phosphorus often limits phytoplankton growth during springtime in the Chesapeake Bay⁴⁸ and drives chl-a variability at one location in Weeks Bay, Alabama U.S.⁴⁹. Furthermore, PO₄³⁻ typically limits growth in freshwater environments⁵⁰, which is characteristic of upstream conditions in estuaries. This dynamic nature of estuaries along the freshwater-to-marine continuum is likely reflected in the modeling results showing that both N and P play a key role in predicting chl-a trends. Interestingly, it was decreasing trends in reduced forms of N (e.g., NH₄⁺) that predicted chl-a trends in this study, which are typically regarded as being more bioavailable (although not necessarily for diatoms⁵¹), and increasing trends in PO₄³⁻ that were correlated with increasing chl-a concentrations. Planktonic primary producers can rapidly take up NH₄⁺, and so the declines in NH₄⁺ concentration may actually reflect increased uptake from increasing primary production, possibly driven by an increase in PO₄³⁻ availability. These nutrient uptake rates may be further increased by coincident rises in temperature and subsequently, phytoplankton growth⁵². Our dataset did not allow us to directly analyze N:P ratios because of numerous instances in which both N and P were below the limit of detection, preventing generation of a meaningful ratio. However, the associations of decreasing NH₄⁺ and increasing PO₄³⁻ trends with chl-a

trends support a claim that nutrient ratios and availability (i.e., stoichiometry) are broadly changing, with important implications for primary productivity, phytoplankton community composition, and food web dynamics⁵³. An important consideration for interpreting our analyses of the putative drivers of chl-a trends is the lack of data describing the organic forms and total pools of N and P. Inorganic nutrients are generally understood to be the most bioavailable compounds for phytoplankton; however, organic N and P can be present at concentrations far exceeding inorganic forms, even by an order of magnitude^{54,55}. Furthermore, many of the organic nutrient forms are indeed bioavailable to different phytoplankton taxa and/or can be recycled into more readily usable forms (i.e., NH₄⁺) at high concentrations upon organic matter decomposition and remineralization⁵⁶⁻⁵⁹. Unfortunately, organic nutrient forms are not consistently measured as part of SWMP at the national scale and could not be included in this analysis. Nonetheless, it is critical to note their potential role in explaining additional variance in trends in chl-a and nutrient cycles and highlight the need for future investigation of the relative importance of organic nutrients to estuarine water quality.

The importance of trends in turbidity for predicting chl-a trends may reflect the increase in phytoplankton biomass and associated organic detritus in suspension. Alternatively, the relative importance of median SpCond (importance value = 0.21; Supplementary Data 5) and precipitation trend (importance value = 0.31; Supplementary Data 5) alongside turbidity could indicate that turbidity trends may indirectly represent altered nutrient and light dynamics resulting from terrestrial inputs and changes in rainfall frequency, duration or intensity, hydrological mixing patterns, or overall

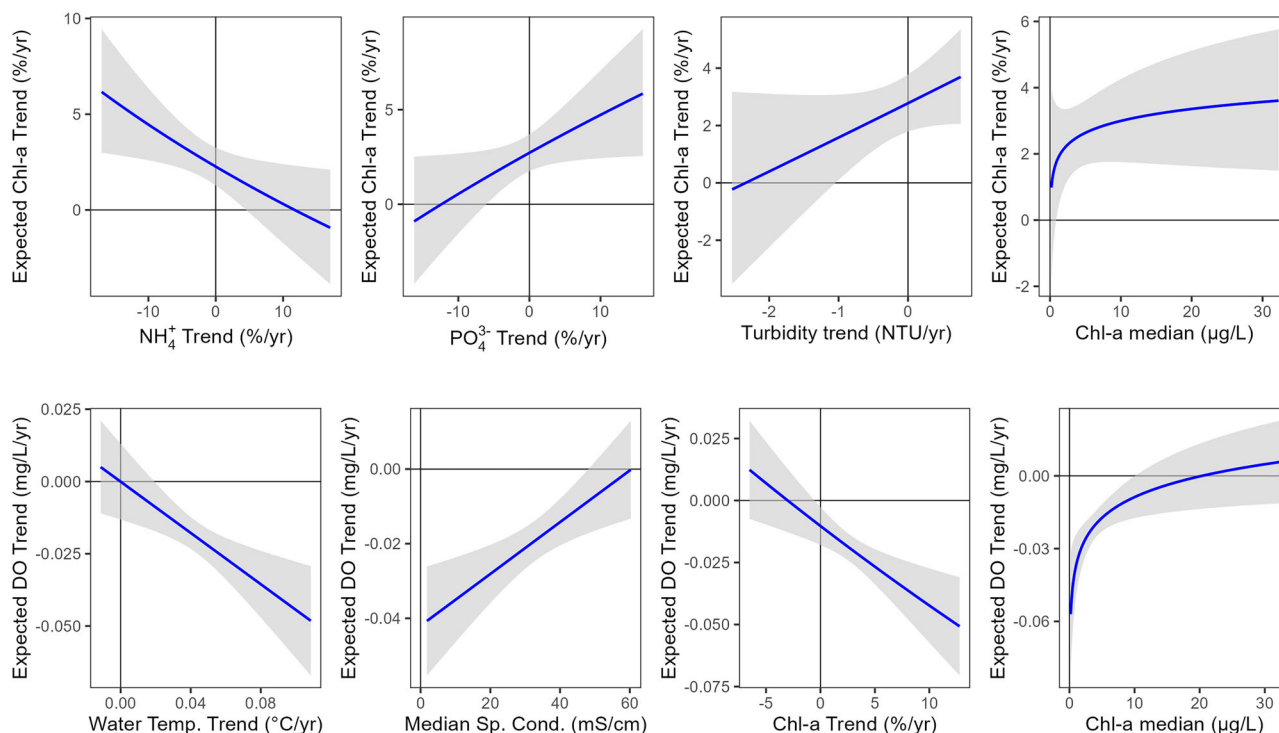


Fig. 6 | Partial effects plots for the top four predictors in the chl-a trend model (top) and dissolved oxygen (DO) trend model (bottom). Each panel shows the predictor effect when all other predictors are held constant at their mean value.

estuarine geomorphology^{60,61}. However, precipitation, SpCond, and turbidity were not highly correlated, indicating that the linkages between these variables are not key drivers of turbidity in this dataset.

The equal weight assigned to trends in chl-a and temperature for predicting changes in DO concentrations could be tied to several complex ecological processes. Temperature directly governs oxygen solubility and is therefore inversely related to DO concentrations; however, solubility alone does not account for changes in DO in this dataset (Supplementary Fig. 9). The importance of chl-a trends in predicting changes in DO (negatively related to DO trend) may result from increased oxygen consumption via accelerated organic matter decomposition, which can in some cases reduce oxygen sufficiently to alter redox conditions and reintroduce dissolved nutrients into the water column that can fuel future phytoplankton growth⁶²; however, our analyses do not indicate that instances of hypoxia are increasing over time. Still, there are some locations where the positive feedback between eutrophication and hypoxia is likely being demonstrated. For example, Old Woman Creek (OWC) NERR in Ohio is not experiencing increasing temperature but is experiencing significant increases in chl-a and decreases in DO, as well as an increasing trend in hypoxia at the Darrow Road site (OWC-DR, Supplementary Data 1). Interestingly, there are decadal nutrient declines in this estuary, and warming trends are mild, which are two main drivers of phytoplankton growth (i.e., chl-a increases) in this study. Climate is expected to have a moderate effect on warming air temperature throughout this region and increase the frequency and magnitude of large, episodic precipitation events⁶³. In the OWC estuary, storm events drive nutrient inputs into the system, as opposed to longer-term increases in nutrient loading⁶⁴. Storm-driven nutrient pulses are not well-represented in our analyses because the SWMP protocol purposely avoids anomalous storm events to build a time series that is representative of average conditions. Thus, in many instances, it is possible that the episodic processes (e.g., storms) that can contribute to long-term change are not well-represented^{65,66}.

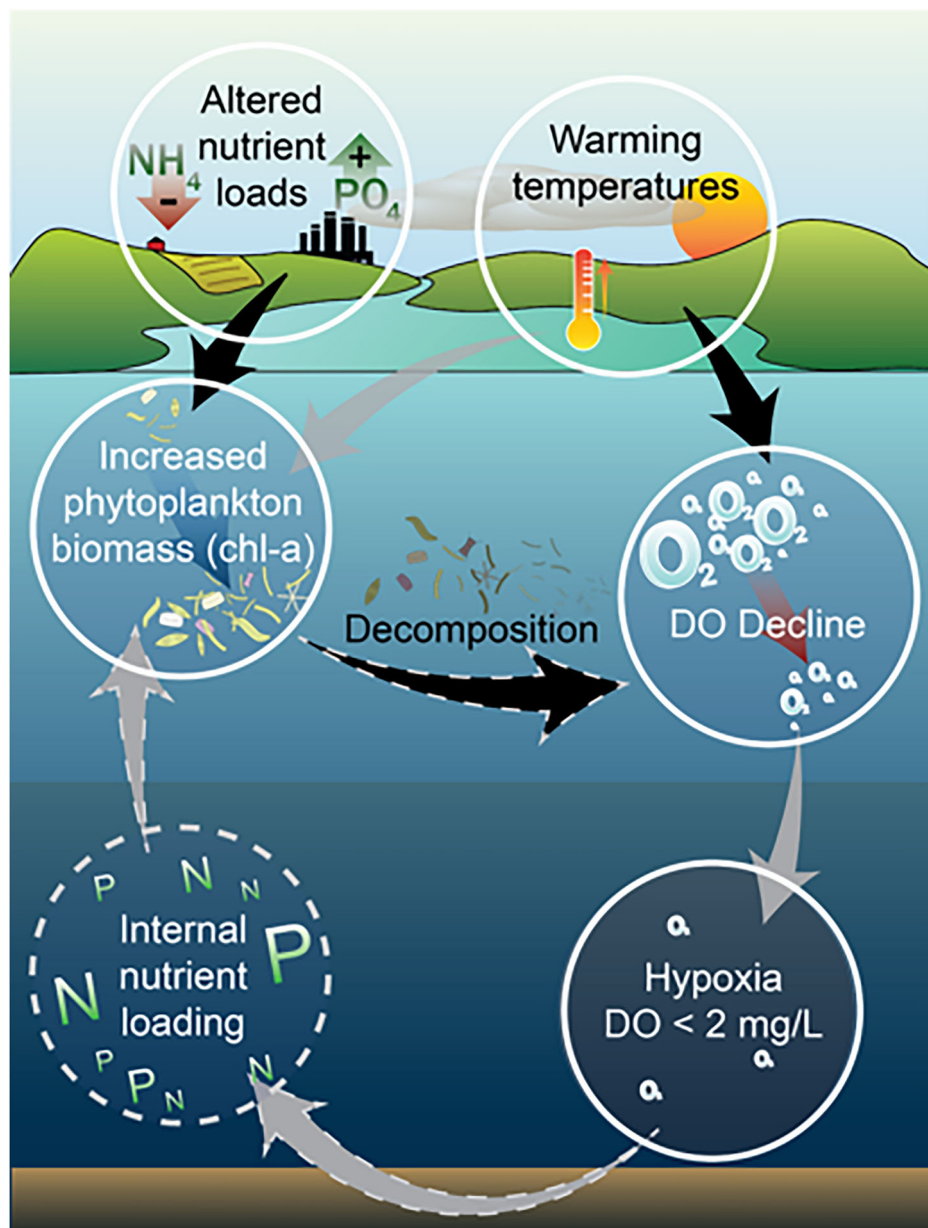
The positive relationship between median SpCond and DO trends (Fig. 5) probably reflects the location of sites along the freshwater-to-marine gradient. Lower SpCond sites experience more freshwater influence, and

these sites are also the most productive, as indicated by median chl-a concentration (Fig. 1), resulting in more oxygen depletion as described above. However, chl-a is more directly linked to phytoplankton biomass rather than primary productivity, which can be controlled by myriad factors such as resource limitation and species-specific responses. There is also a positive relationship between median chl-a (trophic status) and DO and chl-a trends. That is, the highest rates of DO declines are occurring at sites with lowest median chl-a (i.e., the least productive sites), and the highest rates of increase in chl-a are the most productive sites (Fig. 6). This is surprising given the inverse relationship between DO trends and chl-a trends; however, sites with low median chl-a also have the highest rates of temperature increase, suggesting that the linkage between median chl-a and DO trend is actually another signal of temperature effects on DO, as opposed to a relationship with trophic state.

We were unable to elucidate the effects of longer-term cycles such as the Pacific Decadal Oscillation (PDO) and Atlantic Multidecadal Oscillation (AMO), as our dataset is not long enough to capture these processes spanning more than 30 years⁶⁷. However, we acknowledge that these cycles may have significant implications for water quality in estuaries, especially for temperature and salinity (via drought). The PDO is currently in a cool phase since the mid to late 2000s, while the AMO is in a warm phase that began around 1995 and typically lasts 60–80 years. The AMO warm phase is likely a significant contributor to the widespread increase in temperature on sites located on the U.S. East Coast, while the PDO cool phase may be responsible for the nonsignificant trends in temperature in the Pacific Northwest; however, we do still observe some high rates of warming along the West Coast. These direct effects on temperature likely have subsequent effects on DO, as oxygen solubility is strongly tied to temperature, which was also identified as a key driver in our analysis. In addition to impacting SpCond (salinity), drought conditions driven by these cycles may also influence nutrient loading to these systems through changes in runoff delivery as well as biogeochemical cycling of nutrients, but we were not able to test this directly.

The positive feedback between eutrophication and hypoxia is well documented^{7,44}. Generally, increased nutrient inputs lead to a rise in algal

Fig. 7 | Conceptual figure illustrating the positive feedback loop between eutrophication and hypoxia and the linkages that are represented in our modeling results. Black arrows indicate relationships that were supported by modeling results. Gray arrows show processes that were not supported by our analysis, and elements with dashed lines indicate processes that were not explicitly included in our analysis. Some of the symbols are courtesy of the Integration and Application Network (ian.umces.edu/symbols).



biomass (i.e., using chl-a concentration as a proxy). When phytoplankton biomass grows to high enough concentrations, the subsequent die-off and decomposition process can take up large quantities of oxygen, in some cases, reducing oxygen concentration low enough to alter redox conditions in the sediments. This can result in release of dissolved nutrients back into the water column, creating a self-sustaining feedback loop (Fig. 7). The results of our study showed that some of these processes are occurring at sites included in our analysis, but the feedback loop is not closed. For example, our data showed widespread declines in DO but did not show significant changes in the occurrence of hypoxic events, meaning that sustained, large-scale hypoxia-mediated internal loading likely is not occurring. Even so, distributions of median DO concentration (Fig. 1) indicate that concentrations are nearing the typical hypoxia threshold of $<2 \text{ mg L}^{-1}$ and are often below 5 mg L^{-1} which can have important implications for the oxythermal habitat of fish and other organisms⁶⁸.

We have shown that warming temperatures and altered nutrient concentrations spurred a series of changes to multiple dynamically interacting parameters in coastal waters (chl-a, nutrient, and DO concentrations) across the U.S. that were captured by the NERRS SWMP dataset. Altered

nutrient concentrations, not temperature, appear to be the primary driver of increases in chl-a in estuarine systems included in our dataset at this scale, while rising temperatures and algal biomass (chl-a) contribute to declining oxygen concentrations. While we were able to identify broad-scale drivers of chl-a and DO trends, there is likely significant site-specific variability in drivers that accounts for unexplained variance in our analyses. These analyses reinforce that coastal ecosystem dynamics are complex and driven by numerous interacting factors. Future studies that contribute a more holistic approach to nutrient dynamics in estuarine systems will be key. For example, investigations of how shifting nutrient concentrations, forms, and ratios affect plankton size classes, palatability, and toxin production, and consequences for higher trophic levels are needed in a wide range of estuaries with varied biogeochemistry and plankton communities. In addition, studies that specifically examine the role of depth (i.e., stratification vs. well-mixed and related factors including salinity and turbidity) on DO concentration, nutrient cycling, and phytoplankton biomass are needed because stratification can play a key role in nutrient partitioning as well as setting up potentially disparate abiotic (e.g., temperature, salinity) and biotic (e.g., chl-a) conditions. Finally, our results also highlight the importance of

sustained long-term monitoring efforts to disentangle the relative influences of decadal cyclic processes (e.g., the Pacific Decadal Oscillation (PDO) and the El Niño-Southern Oscillation (ENSO)), climate change, and anthropogenic factors on coastal water quality and ecosystem health to reveal underlying long-term trends³³. As coastal systems continue to respond to shifting baselines, only long-term, high-resolution observations will reveal the patterns that emerge over time. Such understanding is not only essential for anticipating ecological change but for guiding adaptive, science-based coastal management in a rapidly transforming world.

Methods

Data aggregation, QA/QC process and parameter selection

Water quality, meteorological, and nutrient data from January 2002 through December 2022 were obtained from the National Estuarine Research Reserve's Centralized Data Management Office (CDMO)⁶⁹ for all SWMP stations, located in 29 estuaries throughout the United States. Only data from currently active (as of 2022) SWMP stations were processed. All data collected as part of the NERRS SWMP adhere to a strict sampling and QA/QC protocol that is available in detail at the CDMO website (<https://cdmo.baruch.sc.edu/data/metadata.cfm>). Specific details on analytical methods, method detection limits (MDL), data flagging procedures, and more are available for all stations in the dataset. We briefly describe some of these methods at a high level in this section below.

Aggregation, compilation, and quality control were performed in R⁷⁰ (v. 4.2.2), using the SWMP⁷¹ (v. 2.5.0) and tidyverse packages⁷². All data flagged by local researchers as “rejected” or “suspect,” following the NERRS SWMP water quality protocol⁴² were excluded from analysis, with exceptions when “suspect” was combined with an “algae bloom” or “instrument recording error; recovered telemetry data” descriptor. Water quality parameters included water temperature, specific conductivity (SpCond; conductance of water, determined by dissolved ions, standardized to 25 °C), DO, turbidity, and pH, while meteorological parameters included precipitation and photosynthetically active radiation (PAR); ultimately, PAR was excluded from all analyses due to changes in instrumentation over time that affected data consistency. We used SpCond, rather than salinity (which is more commonly utilized for estuarine ecosystems), because it is more applicable to the broad range of conditions typically observed in estuarine systems, and salinity is calculated from specific conductivity by the YSI® data loggers as part of the SWMP sampling protocol.

All water quality and meteorological parameters were collected at 15- or 30-min intervals and aggregated to monthly medians, except for data used to calculate the frequency of hypoxic events, which was quantified using 15-min interval data. Months with less than 1 week of data were excluded. Water quality (i.e., not nutrient parameters) data were collected with sensor-based, automated data loggers (YSI® 6-series or EXO2 sondes). Replicate grab samples of surface water were collected for well-mixed sites; from these samples, PO₄³⁻, NO_x, NH₄⁺, and chl-a were determined, and here we utilized the mean value of those replicates. For Hudson NERR, we substituted nitrate for nitrite+nitrate as there were a large number of missing values and these parameters are nearly equivalent in this dataset (Supplementary Fig. 10). All nutrient and chl-a concentration values reported below the MDL were replaced with half the reported MDL. Reserves at higher latitudes may not collect water quality measurements during winter because of significant icing, meaning data points were not available in all months of the year at these stations.

After data aggregation to the monthly scale, each SWMP station was assessed for inclusion in the Principal Component Analysis (PCA) and clustering analysis (i.e., “Status”) and trend analysis. Stations were included in the “Status” analysis (based on PCA and clustering, see below) if they contained >12 months of each of the SWMP water quality, weather, and nutrient parameters listed above (129 stations). These stations were subsequently included in the trend analyses if they contained >10 years of data; however, only 2 of 99 sites had 10–11 years of data, 11 sites had 13–18 years of data, 16 had 18–19 years of data, and 70 sites had 20 years of data or slightly more, up to 21 years (Supplementary Data 4). Three of these long-term

stations were further excluded because their sensors were deployed anomalously far below the water surface (~9, 12, and 20 m; all other long-term sampling sites were at median depths within ~6 m of the surface), leaving 99 “trends” stations. Detailed data processing methodology can be found at https://github.com/Lake-Superior-Reserve/WQ_SWMP_Synthesis/tree/main/R/Data_processing.

Analytical methods: Status

For the 129 sites across 29 reserves described above, we used continuous in situ water quality sensor data (water temperature, pH, SpCond, DO, and turbidity) and monthly nutrient data (chl-a, NO_x, NH₄⁺, and PO₄³⁻) to describe the status of estuarine conditions along U.S. coastlines and identify sites with common characteristics using k-means clustering. The length of these datasets ranged from 2 to 21 years and included all depths down to approximately 19 m, though few are deeper than the 6 m cutoff used in the trend analysis. Site locations encompass the Atlantic, Pacific, Gulf, and Great Lakes coasts and multiple climate zones (including Puerto Rico, Hawaii, and Alaska; Fig. 2). Medians were computed from the previously determined monthly medians for each parameter at each station. The medians for nutrients, chl-a, and turbidity were log-transformed, then mean-centered and standardized to 1 standard deviation. PCA was then applied using the R function `fviz_pca_var()`. Following the Kaiser Criterion⁷³, principal components (PCs) with eigenvalues greater than 1 were retained, which, in this case, led to further analysis of the station scores for the first three PCs.

K-means clustering analysis using the R function `kmeans()` was then conducted on the 129 × 3 PC station scores matrix to highlight groups of stations with similar environmental profiles. K-means clustering is a machine learning algorithm designed to partition a dataset into *K* non-overlapping clusters based on minimizing Euclidean distance within each cluster to its associated centroid⁷⁴. Although unsupervised metrics such as silhouette width suggested setting the *K* number of clusters for this dataset to three, four clusters were selected to capture a fourth, generally eutrophic cluster characterized by lower PO₄³⁻ and NO_x. All PCA outputs and cluster assignments are shown in Supplementary Data 2.

Analytical methods: Trends

For the 99 SWMP stations across 27 reserves with >10 years of data, median values and decadal trends were calculated from monthly medians for all parameters. The sites included in the trend analysis have wide-ranging water quality characteristics with depths ranging from 0.19 to 6.06 m; however, the majority of our sites are shallow, with a median depth of approximately 1.4 m with a standard deviation of 0.3 m. To estimate these trends, we used generalized additive models (GAMs) of the basic form:

$$y \sim T + s(M, k = 12).$$

Here, the trend (*T*) in a given parameter (*y*) was assumed to be linear, and we included a cyclic smoothed term for month (*M*) to account for seasonal patterns in each station-by-parameter time series. The seasonal term had 12 knots or the number of months represented in the data frame for stations that have sondes removed for part of the year due to ice. For nutrient and chl-a parameters, values were natural log-transformed; monthly precipitation was square-root transformed before analysis (for improved residual diagnostics). Autocorrelation of residuals was automatically checked for and if significant autocorrelation was present, the model was re-run to account for it by providing the lag-1 autocorrelation coefficient to the “rho” argument of the `bam()` function. The reported result is the linear trend through time (per year) of a given parameter. For all models we used the `bam()` function within the R package `mgcv`⁷⁵ (version 1.8.41).

To investigate trends in hypoxia, the proportion of time that DO concentration was below 2 mg/L and 5 mg/L was calculated for each month, divided by the number of valid readings. Trends were again calculated within the `bam()` function using the same model as above but using the beta

distribution due to the response being a proportion. In total, we estimated trends for 10 parameters from 99 stations, generating 990 linear slopes.

Analytical methods: Drivers

Trends in chl-a, DO concentration, and proportion of readings where DO < 2 mg/L were chosen as our main response variables because they are common indicators of eutrophication and hypoxia. Median values and trends for water temperature, SpCond, turbidity, chl-a, NH_4^+ , NO_x , PO_4^{3-} , and monthly total precipitation were chosen as our primary predictor variables. Following model fitting, we dropped hypoxia trends as a key response variable because it exhibited little change over time across our geographic domain. Due to collinearity between DO, water temperature, and latitude, the medians (not the trends) of these three variables were combined via PCA, and a station score on PC 1 was used as a predictor in modeling. This PC axis accounted for 89 % of variation in these three collinear parameters. See Supplementary Data 7 for details on each parameter, its units, any transformations, and the distribution used.

To assess how well environmental conditions and trends in environmental conditions explain changes in our key response variables (chl-a trend and DO trend), an information theoretic model selection framework was used⁷⁶. Predictors for each of the three responses were carefully chosen from the prioritized variables described above. No interaction terms were included. All variables were mean-centered and scaled to 1 standard deviation before model-fitting to help with model convergence in mixed models, to enable comparison between standardized coefficients in final models, and to ensure appropriate model selection and averaging^{76–79}. When necessary for interpreting results, coefficients were back-calculated to either their original units or to percent-per-year change (i.e., when log-transformations had been performed).

We used linear mixed effects models to identify possible drivers of chl-a and DO trends (i.e., slope was the model response) and averaged models with $\Delta\text{AICc} < 4$. AIC is the Akaike Information Criterion and AICc is corrected AIC, accounting for instances of small sample sizes (as the sample size increases, AICc converges to AIC). ΔAICc is the difference in AICc scores between the model under consideration and the model with the lowest AICc score (i.e., the “best” model). Global models were constructed with the predictors described above plus a random intercept for reserve. Models were fit using Restricted Maximum Likelihood (REML) with the “lme” function of the R nlme package^{79,80} (v. 3.1.160). To determine whether the random effect (reserve) was necessary to include in our models, a simple linear model without a random effect was constructed via nlme::gls(), also using REML, which is required when using AIC to compare models with different random effect structures⁸¹. The models with and without a random effect were compared via AICc, a modification of AIC that corrects for small sample size. When the model without the random effect was within 2 AICc of the model with it, the random effect was dropped from the global model (ultimately, a random effect was only present for the chl-a model). When the random effect was required (e.g., chl-a models), the global model was re-fit in “lme” using Maximum Likelihood (ML) for subsequent model selection. When the random effect was not required (e.g., DO trend models), global models were re-fit as simple linear models, using stats::lm().

All model selection, averaging, and variable importance calculations were performed using the R MuMIn package⁸² (v. 1.47.5). All-subsets selection was conducted using the dredge() function. This resulted in 65,536 candidate models for predicted DO trends and 32,768 candidate models for predicting trends in chl-a. Due to varying recommendations on AIC thresholds for top model sets^{77,83,84} and based on the number of models that would be included using different thresholds⁷⁷, top model sets were generated for each response based on ΔAICc for thresholds of 2, 4, and 6 AICc units. Results using $\Delta\text{AICc} < 4$ are presented here, as this cutoff was quantitatively similar to the commonly used threshold of 2 while including models that are still plausible, increasing the pool of candidate models used in model averaging. Final standardized coefficients were generated by averaging the top model sets with model.avg(), using the full-model method, which is appropriate for comparing relative effect sizes when there is high

model uncertainty⁷⁸. Variable importance was assessed partly by comparing standardized coefficients but also by using the ‘relative importance’ metric calculated by the sw() function. The relative importance is the sum of Akaike weights of models in which a predictor appears. It ranges from 0–1 and can be interpreted as the probability that the predictor is in the “best” model^{77,78}. We also report how many models from the top set a predictor appeared in. Predictors that are included in highly weighted and/or a high number of models will have higher weights than those in few and/or low-weighted models. Graphics were generated using the ggplot2 package in R⁷².

Data availability

The SWMP data this study used are available to query and download from the National Estuarine Research Reserve System’s Centralized Data Management Office at <https://cdmo.baruch.sc.edu/>.

Code availability

All custom scripts developed for this study are available via the GitHub repository associated with the NERR SWMP Synthesis project at https://github.com/Lake-Superior-Reserve/WQ_SWMP_Synthesis/tree/main/R.

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Author contributions

K.L.R.: Conceptualization, project administration, methodology, validation, formal analysis, investigation, writing—original draft, writing—review and editing, visualization; R.P.D.: Conceptualization, project administration, methodology, validation, formal analysis, investigation, writing—original draft, writing—review and editing. K.A.C.: Methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft, writing—review and editing, visualization; J.C.G.: Writing—original draft, writing—review and editing; D.M.S.: Conceptualization, methodology, writing—review and editing; J.L.K.: Conceptualization, methodology, software, investigation, writing—original draft, writing—review and editing; A.R.H.: Conceptualization, methodology, writing—original draft, writing—review and editing, H.N.N.: Conceptualization, methodology, writing—original draft, writing—review and editing; S.E.M.: Writing—original draft, writing—review and editing; N.G.D.: Conceptualization, writing—original draft, writing—review and editing; J.T.R.: Conceptualization, writing—original draft, writing—review and editing, visualization; J.L.D.: Conceptualization, methodology, writing—original draft, writing—review and editing; T.J.C.: Conceptualization, methodology, writing—original draft, writing—review and editing; C.R.P.: Writing—original draft, writing—review and editing; C.T.F.: Conceptualization, methodology, writing—original draft, writing—review and editing; C.K.: Conceptualization, writing—original draft, writing—review and editing; D.B.P.: Conceptualization, validation, formal analysis, writing—original draft, writing—review and editing; T.M.G.: Conceptualization, methodology, writing—original draft, writing—review and editing.

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Competing interests

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Additional information

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Correspondence and requests for materials should be addressed to Kaitlin L. Reinl.

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¹Department of Biological Sciences, Rensselaer Polytechnic Institute, Troy, NY, USA. ²North Inlet-Winyah Bay National Estuarine Research Reserve, Baruch Marine Field Laboratory, University of South Carolina, Georgetown, SC, USA. ³Ecological Dynamics LLC, Greensboro, NC, USA. ⁴Catbird Stats LLC, Gautier, MS, USA. ⁵Waquoit Bay National Estuarine Research Reserve, Massachusetts Department of Conservation and Recreation, Falmouth, MA, USA. ⁶Grand Bay National Estuarine Research Reserve, Moss Point, MS, USA. ⁷Coastal and Marine Extension Program, Mississippi State University, Biloxi, MS, USA. ⁸Chesapeake Bay National Estuarine Research Reserve in Virginia, Virginia Institute of Marine Science, Gloucester, VA, USA. ⁹Wells National Estuarine Research Reserve, Wells, ME, USA. ¹⁰South Slough National Estuarine Research Reserve, Oregon Department of State Lands, Charleston, OR, USA. ¹¹NOAA Office for Coastal Management, Silver Spring, MD, USA. ¹²Lake Superior National Estuarine Research Reserve, University of Wisconsin Madison, Division of Extension, Madison, WI, USA. ¹³North Carolina Coastal Reserve and National Estuarine Research Reserve, North Carolina Department of Environmental Quality Division of Coastal Management, Beaufort, NC, USA. ¹⁴ACE Basin National Estuarine Research Reserve, South Carolina Department of Natural Resources, Charleston, SC, USA. ¹⁵Old Woman Creek National Estuarine Research Reserve, Ohio Department of Natural Resources, Huron, OH, USA. ¹⁶Guana Tolomato Matanzas National Estuarine Research Reserve, Florida Department of Environmental Protection, Ponte Vedra Beach, FL, USA. ¹⁷University of North Florida, Jacksonville, FL, USA. ¹⁸Jacques Cousteau National Estuarine Research Reserve, Rutgers University, Tuckerton, NJ, USA. ¹⁹Great Bay National Estuarine Research Reserve, New Hampshire Fish and Game, Greenland, NH, USA. ✉ e-mail: reinlk@rpi.edu